

## Chapter 1

# INTRODUCTION

Heating or cooling of materials is an indispensable part of processing, production, fabrication or shopfloor jobs in engineering practice. Hardly any job or operation can be identified that does not involve heating or cooling of a body or a material at some stage or other. Whenever a material or a fluid stream has to be heated or cooled, or whenever a material has to be stored in a hot or cold condition, we have to see that it is done in the most economic way possible so that the loss or gain of heat during the process is minimum.

Because of the ubiquity of heat transfer phenomena in industrial processes, its study has found an important place in engineering curricula. An engineer should have a sound understanding of the basic principles and applications of heat transfer. How do we *define* heat transfer? Heat transfer is the transport of heat energy from one point in a medium to another or from one medium to another in the presence of a temperature gradient or a temperature difference. The temperature difference between two points in the same medium, or between two mediums which are in thermal contact, is known as the *driving force* for heat transfer. The subject of heat transfer involves a detailed study of the physical mechanism of transport of heat energy, the methods of calculation or prediction of rates of transport of heat energy in various practical situations, and the applications of the theoretical principles to sizing, design and improvement of related equipment used in practice.

### 1.1. MODES OF HEAT TRANSFER

Three modes of heat transfer are recognized: conduction, convection and radiation. The first two modes appear to be dominant in many practical fields. However, radiation is the most important mode of heat transfer at high temperatures.

#### 1.1.1. Conduction

Conduction heat transfer is an atomic or molecular process. It occurs in the presence of a temperature difference and is not accompanied by any macroscopic or bulk motion in the medium. Conduction is the only mode of heat transfer in a solid medium. It may also occur in a stagnant liquid, or a gaseous medium. The basic law of conduction, *the Fourier's law*, is explained in Chapter 2, and a few situations of steady state conduction are analyzed. Conduction heat transfer plays a major role in heat loss from furnaces, hot pipe lines (e.g. steam pipes), and process vessels and equipment. It is also important in the process of heat gain by a fluid stored in an insulated tank at a cryogenic temperature. In fact, conduction has a role to play in virtually all of the heat transfer equipment, but in many cases it may not have a governing role.

#### 1.1.2. Convection

Existence of motion or a velocity field in a liquid or a gaseous medium greatly enhances the rate of heat transfer. Convection means the transport of heat energy by way of displacement of fluid elements from one point to another point which is at a different temperature. Convection may be of two types *forced convection and free or natural convection*. Forced convection occurs when motion in the medium is caused by an external mechanical agency such as a pump, a blower, an agitator, etc. or by an externally imposed pressure gradient. Free or natural convection occurs when motion in the medium is created by an adverse density gradient, as a result of temperature difference. This happens if the temperature of a fluid at a lower level becomes higher than that at an upper level.

While calculation of the rate of heat transfer by conduction is based on the Fourier's law, the effect of fluid motion on convective heat transfer rate cannot be taken into account by using any similar law. An empirical *heat transfer coefficient*, which is based on the *phenomenological observation that heat flux is proportional to the temperature driving force*, is defined instead. The major component of the study of convective heat transfer involves the development and application of correlations for the heat transfer coefficient in diverse physical situations.

### 1.1.3. Radiation

A body at a temperature above absolute zero always emits energy in the form of electromagnetic waves. The rate of release of such energy is proportional to the fourth power of the absolute temperature of the body. This phenomenon is called radiation and the basic governing law is known as the *Stefan-Boltzmann law*. Ordinarily, the contribution of the radiative component to the total rate of heat transfer from a body becomes significant, if the temperature of the body is sufficiently high. All bodies, however, are not equally good emitters of radiation. A standard or perfect emitter is called a *blackbody*, and acts as a reference with which any other body may be compared in respect of its effectiveness as an emitter. This property of a body is called its *emissivity*. Another quantity which comes into play while calculating the rate of radiative heat exchange between two bodies is called the *view factor*. It takes into account the fraction of the area of a body which can be seen by another body. The study of radiative heat transfer involves the basic principles of radiation and their applications to computations of heat transfer rates in different geometrical configurations of physical systems.

In systems of practical interest, heat transfer mostly occurs by a combination of two or all of the above three modes. Consider, for example, the case of heat loss from a furnace wall. The inner surface of the wall gains heat from the flame and the combustion gases by both radiation and convection. Heat transfer through the refractory and the insulating brick layers of the furnace wall occurs by conduction. Finally, the transfer of heat from the outer surface of the furnace to the ambient occurs by convection and radiation. Boiling of liquids and condensation of vapours, which involve a change of phase, are other ramifications of heat transfer that occur mainly by convection.

## 1.2. HEAT TRANSFER EQUIPMENT

Heat transfer equipment is designed and built by utilizing the basic principles and correlations of heat transfer along with other relevant physical and engineering principles. The objective is to achieve the desired rate of heat transfer from one medium to another in the best possible way. Most of the heat transfer equipment comes under the general name of *heat exchangers*, and includes *reboilers*, *condensers*, *vaporizers* as well. An engineer, particularly a chemical or a mechanical engineer, must have a basic understanding of the design methodology, construction and operation of heat exchangers.

## 1.3. SYSTEMS OF MEASUREMENT, UNITS AND DIMENSIONS

In this section we give a brief overview of the *units* and *dimensions* of physical quantities and also of the interrelations among the different units used for measurement of such quantities. However, we will generally use the SI units throughout this book.

A *unit* is a standard of measurement of a physical quantity. *Dimension* of a quantity indicates how a physical quantity is related to the basic or fundamental quantities. A system of measurement is based on a few fundamental or basic quantities and their units, which are called the “basic units”. All other quantities are obtainable by some kind of combination of the basic

quantities, and are called “derived quantities”. The unit of a derived quantity is obtainable in terms of the basic units in any system of measurement.

Until recently, the CGS (Centimetre-Gram-Second) and the FPS (Foot-Pound-Second) systems were most widely used, particularly in scientific literature. The fundamental quantities and their dimensions in these systems are shown in Table 1.1. A few derived quantities of both these systems are also shown in the table. In the study of heat transfer, we will consider these four fundamental quantities-mass, length, time and temperature as the basic units.

**Table 1.1** Relevant fundamental and derived quantities of the CGS and the FPS systems

| <i>Quantity</i>    | <i>Dimension</i>    |                     | <i>Unit</i>                   |                                   |
|--------------------|---------------------|---------------------|-------------------------------|-----------------------------------|
|                    | CGS                 | FPS                 | CGS                           | FPS                               |
| <b>Fundamental</b> |                     |                     |                               |                                   |
| Mass               | M                   | M                   | g                             | lb                                |
| Length             | L                   | L                   | cm                            | ft                                |
| Time               | t                   | t                   | s                             | s                                 |
| Temperature        | T                   | T                   | °C                            | °F                                |
| <b>Derived</b>     |                     |                     |                               |                                   |
| Area               | L <sup>2</sup>      | L <sup>2</sup>      | cm <sup>2</sup>               | ft <sup>2</sup>                   |
| Density            | M L <sup>-3</sup>   | M L <sup>-3</sup>   | g/cm <sup>3</sup>             | lbm/ft <sup>3</sup>               |
| Velocity           | L t <sup>-1</sup>   | L t <sup>-1</sup>   | cm/s                          | ft/s                              |
| Force              | M L t <sup>-2</sup> | M L t <sup>-2</sup> | g cm/s <sup>2</sup><br>(dyne) | lb ft/s <sup>2</sup><br>(poundal) |

### 1.3.1. English Engineering System

This system of measurement and units has been largely used in engineering calculations all over the world (Table 1.2). It differs from the PFS system in that force is considered as a fundamental quantity here whereas in the PFS system, force is a derive quantity (see Table 1.1). So this system uses fives fundamental quantities (i.e. mass, length, time, temperature and force, leaving aside the quantities used for electrical measurements).

**Table 1.2** Units and dimensions in the English Engineering System

| <i>Quantity</i>    | <i>Dimension</i>                  | <i>Unit</i> |
|--------------------|-----------------------------------|-------------|
| <b>Fundamental</b> |                                   |             |
| Mass               | M                                 | lbm         |
| Length             | L                                 | ft          |
| Time               | t                                 | s           |
| Temperature        | T                                 | °F          |
| Force              | F                                 | lbf         |
| <b>Derived</b>     |                                   |             |
| Energy             | F L                               | ft-lbf      |
| Viscosity          | M L <sup>-1</sup> t <sup>-1</sup> | lbm/ft s    |

### 1.3.2. The International System (SI) of Units (The Syst me Internationale (SI) de Unit s)

The International System of units was recommended in 1960 at the Eleventh General Conference on Weights and Measures. Since then, it has gradually replaced the English system beginning from the late sixties. Nevertheless, the English system is still preferred by some professionals, because many design data, charts and correlations are readily available in this system.

Some important basic units of SI and also a few derived units are listed in Table 1.3. Conversion of the unit of a quantity from the English system to SI and vice versa, is often required in engineering calculations. Conversion of some of the quantities relevant to our purpose is given in Table 1.4. In engineering practice relating to heat transfer, the quantities “kcal”, “m”, “h” and “°C” are commonly used.

**Table 1.3** Relevant units and dimensions in the SI

| <i>Quantity</i>                    | <i>Dimension</i>                               | <i>Unit</i>                                                                    |
|------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Fundamental or basic units*</b> |                                                |                                                                                |
| Mass                               | M                                              | kg                                                                             |
| Length                             | L                                              | m                                                                              |
| Time                               | t                                              | s                                                                              |
| Temperature                        | T                                              | K (kelvin)                                                                     |
| <b>Derived units</b>               |                                                |                                                                                |
| Force                              | M L t <sup>-2</sup>                            | $\frac{\text{kg m}}{\text{s}^2}$ = newton (N)                                  |
| Work                               | M L <sup>2</sup> t <sup>-2</sup>               | $\frac{\text{kg m}^2}{\text{s}^2}$ = N m = joule (J)                           |
| Specific heat                      | L <sup>2</sup> t <sup>-2</sup> T <sup>-1</sup> | $\frac{\text{kg m}^2}{\text{s}^2 \text{ kg K}} = \frac{\text{J}}{\text{kg K}}$ |
| Power                              | M L <sup>2</sup> t <sup>-3</sup>               | $\frac{\text{kg m}^2}{\text{s}^3}$ = watt, W (J/s)                             |

\* The other fundamental or basic units of SI are:  
electric current: ampere (A); amount of substance: mole (mol); and luminous intensity: candela (cd).

## 1.4. EXAMPLES OF UNIT CONVERSION

(i) The coefficient of viscosity of water is 1 cP at 20°C. Calculate its value both in English and SI units.

$$1 \text{ cP} = 0.01 \text{ poise} = 0.01 \frac{\text{g}}{\text{cm s}} = \frac{(0.01)(10^{-3} \text{ kg})}{(10^{-2} \text{ m})(\text{s})} = 0.001 \frac{\text{kg}}{\text{m.s}} \text{ in SI unit.}$$

$$1 \text{ cP} = 0.01 \text{ poise} = 0.01 \frac{\text{g}}{\text{cm s}} = \frac{(0.01) \left( \frac{1}{453.6} \text{ lbm} \right)}{\left( \frac{1}{30.48} \text{ ft} \right) \left( \frac{1}{3600} \text{ h} \right)} = 2.42 \frac{\text{lbm}}{\text{ft h}}$$

(ii) The thermal conductivity of magnesite at 600°C is  $6.7 \times 10^{-3} \text{ cal/cm s } ^\circ\text{C}$ . Find its value both in the English and SI units.

$$6.7 \times 10^{-3} \frac{\text{cal}}{\text{cm s } ^\circ\text{C}} = \frac{(6.7 \times 10^{-3}) \left( \frac{1}{252} \text{ Btu} \right)}{\left( \frac{1}{3600} \text{ h} \right) \left( \frac{1}{30.48} \text{ ft} \right) \left( \frac{9}{5} ^\circ\text{F} \right)} = 1.62 \frac{\text{Btu}}{\text{h ft } ^\circ\text{F}}$$

$$6.7 \times 10^{-3} \frac{\text{cal}}{\text{cm s } ^\circ\text{C}} = \frac{(6.7 \times 10^{-3})(4.187 \text{ W s})}{(\text{s}) \left( \frac{1}{100} \text{ m} \right) (\text{K})} = 2.805 \frac{\text{W}}{\text{m K}}$$

**Table 1.4** Units and conversions of some important physical quantities

| Quantity                  | SI                                                                                                | English                    | CGS                         | SI $\rightarrow$ English                                                                                                                     | English $\rightarrow$ SI                                                           | Other units/conversion                                                                                                                              |
|---------------------------|---------------------------------------------------------------------------------------------------|----------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Mass                      | kg                                                                                                | lbm                        | g                           | 1 kg = 2.2046 lbm                                                                                                                            | 1 lbm = 0.4536 kg                                                                  | 1 lbm = 453.6 g                                                                                                                                     |
| Length                    | m                                                                                                 | ft                         | cm                          | 1 m = 3.2808 ft                                                                                                                              | 1 ft = 0.3048 m                                                                    | 1 ft = 30.48 cm                                                                                                                                     |
| Area                      | m <sup>2</sup>                                                                                    | ft <sup>2</sup>            | cm <sup>2</sup>             | 1 m <sup>2</sup> = 10.7639 ft <sup>2</sup>                                                                                                   | 1 ft <sup>2</sup> = 0.0929 m <sup>2</sup>                                          | 1 ft <sup>2</sup> = 929 cm <sup>2</sup>                                                                                                             |
| Volume                    | m <sup>3</sup>                                                                                    | ft <sup>3</sup>            | cm <sup>3</sup>             | 1 m <sup>3</sup> = 35.3134 ft <sup>3</sup>                                                                                                   | 1 ft <sup>3</sup> = 0.028317 m <sup>3</sup>                                        | 1 ft <sup>3</sup> = 28317 cm <sup>3</sup>                                                                                                           |
| Density                   | kg/m <sup>3</sup>                                                                                 | lbm/ft <sup>3</sup>        | g/cm <sup>3</sup>           | 1 kg/m <sup>3</sup> = 0.06243 lbm/ft <sup>3</sup>                                                                                            | 1 lbm/ft <sup>3</sup> = 16.018 kg/m <sup>3</sup>                                   | 1 lbm/ft <sup>3</sup> = 16018 g/cm <sup>3</sup>                                                                                                     |
| Force                     | N (kg m/s <sup>2</sup> )                                                                          | lbf                        | dyne (= 10 <sup>-5</sup> N) | 1 N = 0.2248 lbf                                                                                                                             | 1 lbf = 4.4482 N                                                                   | 1 lbf = 4.4482 $\times$ 10 <sup>5</sup> dyne                                                                                                        |
| Pressure                  | N/m <sup>2</sup><br>(= pascal, Pa)<br>(= 1.4504 $\times$ 10 <sup>-4</sup> lbf/inch <sup>2</sup> ) | lbf/ft <sup>2</sup>        | dyne/cm <sup>2</sup>        | 1 N/m <sup>2</sup> = 1 Pa<br>(= 2.0885 $\times$ 10 <sup>-2</sup> lbf/ft <sup>2</sup><br>= 6.8946 $\times$ 10 <sup>3</sup> N/m <sup>2</sup> ) | 1 lbf/ft <sup>2</sup> = 47.881 N/m <sup>2</sup><br>1 lbf/inch <sup>2</sup> = 1 psi | 1 atm = 14.7 psi<br>= 1.013 $\times$ 10 <sup>5</sup> N/m <sup>2</sup><br>= 1.013 bar = 1.033 kg cm <sup>-2</sup><br>1 kg/cm <sup>2</sup> = 0.97 atm |
| Power                     | W                                                                                                 | ft-lbf/s                   | dyne-cm/s                   | 1 W = 0.737 ft-lbf/s<br>= 1.34 $\times$ 10 <sup>-3</sup> hp                                                                                  | 1 hp = 746 W<br>= 641.6 kcal/h                                                     |                                                                                                                                                     |
| Heat                      | J (= N m)                                                                                         | Btu                        | cal                         | 1 J = 9.4783 $\times$ 10 <sup>-4</sup> Btu                                                                                                   | 1 Btu = 1055.04 J                                                                  | 1 cal = 4.187 J (or W s)<br>1 Btu = 252 cal                                                                                                         |
| Heat flux                 | W/m <sup>2</sup>                                                                                  | Btu/(h ft <sup>2</sup> )   | cal/(s cm <sup>2</sup> )    | 1 W/m <sup>2</sup> = 0.317 Btu/(h ft <sup>2</sup> )                                                                                          | 1 Btu/(h ft <sup>2</sup> ) = 3.154 W (or J/s)                                      | 1 Btu/(h ft <sup>2</sup> )<br>= 2.712 kcal/(h m <sup>2</sup> )                                                                                      |
| Heat flow rate            | W                                                                                                 | Btu/h                      | cal/s                       | 1 W = 3.4121 Btu/h                                                                                                                           | 1 Btu/h = 0.2931 W                                                                 | 1 Btu/h = 0.252 kcal/h                                                                                                                              |
| Heat transfer coefficient | W/(m <sup>2</sup> K)                                                                              | Btu/(h ft <sup>2</sup> °F) | cal/(s cm <sup>2</sup> °C)  | 1 W/(m <sup>2</sup> K) = 0.1761 Btu/(h ft <sup>2</sup> °F)                                                                                   | 1 Btu/(h ft <sup>2</sup> °F) = 5.678 W/(m <sup>2</sup> K)                          | 1 Btu/(h ft <sup>2</sup> °F)<br>= 4.88 kcal/(h m <sup>2</sup> °C)                                                                                   |
| Specific heat             | J/(kg K)                                                                                          | Btu/(lbm °F)               | cal/(g °C)                  | 1 J/(kg K)<br>= 2.3884 $\times$ 10 <sup>-4</sup> Btu/(lbm °F)                                                                                | 1 Btu/(lbm °F) = 4187 J/(kg K)                                                     |                                                                                                                                                     |
| Thermal conductivity      | W/(m K)                                                                                           | Btu/(h ft °F)              | cal/(s cm °C)               | 1 W/(m K) = 0.5778 Btu/(h ft °F)                                                                                                             | 1 Btu/(h ft °F) = 1.7307 W/(m K)                                                   | 1 Btu/(h ft °F)<br>= 1.488 kcal/(h m °C)                                                                                                            |
| Viscosity                 | kg/(m s)<br>(= Pa s)                                                                              | lbm/(ft h)                 | g/(cm s)<br>(= poise)       | 1 kg/(m s) = 2419.2 lbm/(ft h)                                                                                                               | 1 lbm/(ft h) = 4.133 $\times$ 10 <sup>-4</sup> kg/(m s)                            | 1 poise = 2.42 lbm/(ft h)<br>= 0.1 kg/(m s)                                                                                                         |

*Note:* The brackets in the denominators of the above units have been dropped in subsequent chapters for ease of convenience.